

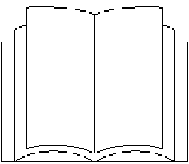


Operator's manual for Attachment



Flour / grain dispensing bucket

Product number: 300-700-series loaders: A1194



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. Original instructions.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even users with only little experience can use the attachment and the loader safely. The manual also includes important instructions for experienced AVANT operators. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT Flour/grain dispensing bucket is an attachment designed and manufactured to suite AVANT multi purpose loaders for distributing flour, crushed grain or similar materials to cattle feeding table. The bucket can be filled from the ground or from a separate silo. Dispensing is carried out with a feeding auger and the discharge is at the right side of the bucket. Upon order the bucket is available with discharge to the left. The bucket can also be equipped with a larger feeding auger to enable it to dispense mash or similar material.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator, but certain more demanding tasks must be left to qualified professionals. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the flour/grain dispensing bucket

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
 - monitor and prevent any hazardous situations
 - keep the equipment intact and ensure a long service life
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- When mounting the attachment to the loader, ensure that the locking pins of the loader's quick attach plate are in the lower position and that they **have locked the bucket to the loader**.
 - When filling the bucket from a silo keep it securely against the ground and attached to the loader.
 - The loader and its controls are to be used by only one operator at a time. Do not go near the attachment if another person is operating the controls of the loader, or allow others near the machine when operating the loader.
 - Always transport the bucket as low and as close to the machine as possible to keep the centre of gravity low.
 - Pay attention to the surroundings and any other persons and machines moving in the vicinity. When transporting the attachment, pay attention to the contours of the terrain and other hazards, such as branches and trees reaching the driver's seat.



- When taking the bucket off from the loader, ensure that it is secured against accidental movement. To prevent the attachment from tipping over, empty it first and lower it on even surface. Ensure the proper support of bucket during storage periods.
- Shut down the loader engine before any maintenance or adjustments. Ensure that the attachment is properly supported during maintenance, inspections or cleaning.
- Never go under a raised attachment or loader boom.
- Be sure not to tilt the bucket too far up to prevent material from spilling from the bucket.
- Before starting the dispenser bucket, make sure that there are no foreign objects in the bucket that could be ejected from the attachment or that could damage the equipment.
- Do not modify the structure of the attachment in a manner that would affect its safety. Use the flour/grain dispenser only for its intended use.
- Check the auger regularly. Read the instructions on maintenance and servicing and only use original spare parts.
- Also read the safety instructions for the loader in the loader manual.



Improper or careless use of the equipment can cause hazardous situations. Use the bucket only to dispense materials that are suitable for cattle.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or to the danger area between the boom and the attachment. Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.



Never leave the machine unattended while the engine is running.

Remember to wear proper protective clothing:



- Never wear loose clothing, long scarves, jewellery or other items that might become entangled in rotating parts of the dispenser bucket. If needed, wear a hat to prevent long hair from coming into contact with the rotating parts.



- The noise level in the driver's seat may exceed 85 dB(A). Wear hearing protection while working with the loader.



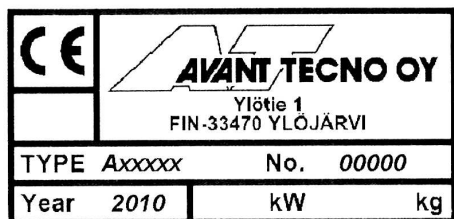
- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.

4. Flour/grain dispensing bucket specifications

Listed on the following page are the markings and labels fastened to the flail mower, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.



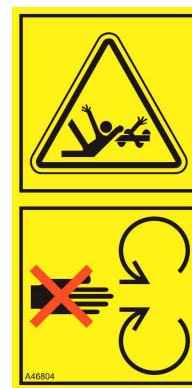
Identification plate of the attachment



A46771



A46772



A46804

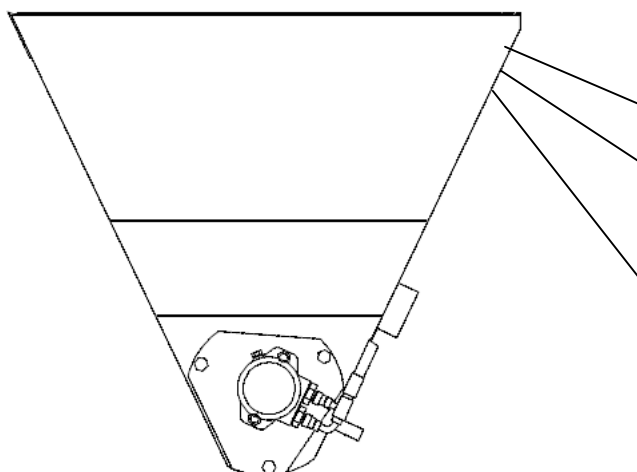
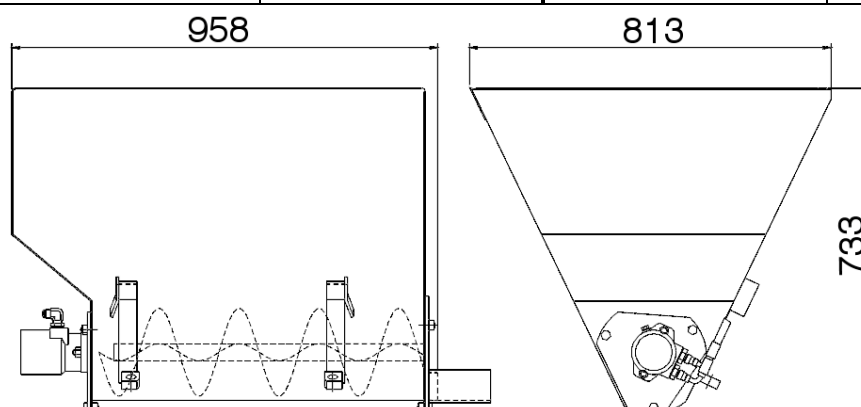


Table 1 – Decal warning messages and locations

Decal	Explanation
A46771	Read instructions before use.
A46772	Do not go under a raised attachment or the loader boom, keep a safe distance to lifted equipment.
A46806	Entanglement hazard, keep clear of moving parts, do not leave the equipment running. Operate the attachment only from the driver's seat. Beware also of entanglement of clothes, jewellery or long hair.

Table 2 – Flour/Grain dispensing bucket technical data

Product number: A1194			
Working width:	920 mm	Volume:	300 l
Total width:	1 100 mm	Weight:	90 kg
Height:	700 mm	Maximum input of hydraulic energy:	70 l/min 22,5 MPa (225 Bar)
Length:	810 mm	Suitable AVANT loader*	300–700



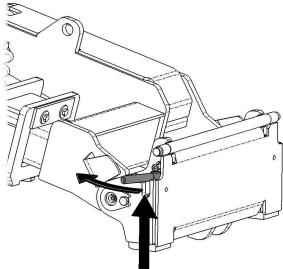
5. Assembly of the dispensing bucket

The dispensing bucket is ready for use as delivered. Mount the attachment and its hydraulic hoses to the loader as instructed and practise the use of the equipment at a safe area. Read the instructions related to installation, using, and maintenance before operating the equipment

5.1. Attaching the bucket to the loader

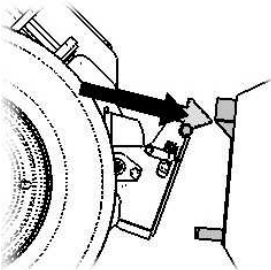
Attaching the attachment to the loader is quick and easy, but it must be done carefully. If the tool is not locked to the loader, it may come off from the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.

The attachment is installed to the loader boom by using the quick attach plate and the counterpart on the attachment. The locking pins are semi-automatic; they will lock when the attachment is tilted towards the loader at low height. The attachment is mounted to the loader as follows:



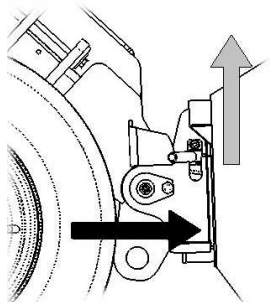
Step 1:

- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.

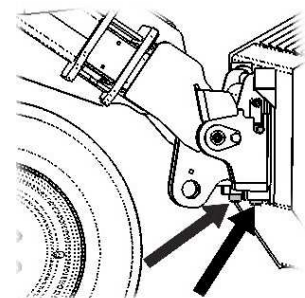


Step 3:

- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins.



Unlocked attachment may fall on the boom or operator and cause injuries. Before moving or lifting the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Never move or raise an attachment that has not been locked.





Do not use the automatic locking of the locking pins **at a height of over one metre.**

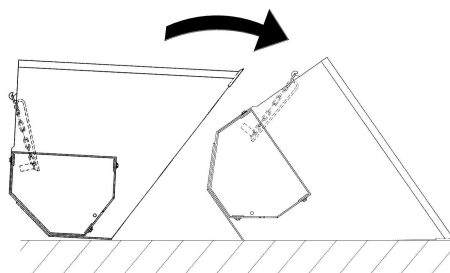
If the locking pins do not return to the normal position when the front plate is tilted, do not lift the attachment more. Lower the attachment on the ground and secure the locking manually.

5.2. Storage of the dispenser bucket

To ensure the stability of the bucket during storage, use it until it is empty or empty it on purpose before taking it off from the loader. The bucket should be stored so that its longest side is against the ground surface, as shown in figure below.



Unattached bucket may tip over; keep a safe distance. Ensure the stability by placing the attachment firmly on the ground on stable position.



Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example. Do not leave the attachment to a position where water will stay in the bucket and prevent water from getting in the bucket for example by placing a suitable block of wood under the attachment. Before longer storage periods clean the bucket thoroughly, touch up the paint where needed, and lubricate the bearing to prevent rust damage.

5.3. Connecting and disconnecting the hydraulic fittings

The attachment is equipped with either standard quick couplings or the multiconnector system. If you want to change the type of the hydraulic couplings, contact your Avant dealer or service point for instructions or installation services.



Never connect or disconnect quick couplings or other hydraulic components if the control lever of the auxiliary hydraulics control lever is locked on or while the system is pressurized. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead to unintended movements of the attachment or ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

Connecting standard quick couplings:

Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there will be no residual pressure in the system and it is possible to connect the fittings.

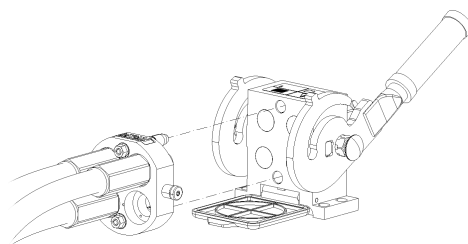
To connect and disconnect the standard couplings, move the collar at the end of the female fitting. The hoses should be connected so that the coupling equipped with a coloured cap is connected to the corresponding coupling of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. When disconnecting the standard quick couplings a small amount of oil may drip from the couplings. Wear protective gloves and have some cloth at hand to keep the equipment clean.



Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down. Connect and lock the multiconnector by turning the lever towards the loader.

The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Also shut down the loader and release the residual hydraulic pressure.



Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface. Turn the control lever of the auxiliary hydraulics to its neutral position.



When taking the attachment off from the loader, always disconnect the hydraulic couplings before unlocking the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the auxiliary hydraulics control lever of the loader, when the engine is turned off. To make sure that there will not be residual pressure in the hydraulic system of the attachment, shut down the loader engine and move the auxiliary hydraulics control lever of the loader before disconnecting the couplings.

6. Instructions for use

Check the attachment and the operating environment once more before starting the machine. Quick inspection of the dispenser and the operating environment prior to use are a part of ensuring safety and the best performance of the equipment.



Never go under a raised boom or bucket component. Note that the loader boom moves down even if the engine has been stopped.



Never perform any adjustments, maintenance or cleaning of the equipment if the bucket can start by accident. Shut down the loader engine first.

- Check that the dispenser bucket is intact and that the feeding screw can rotate freely and that there is no string or other foreign material wound around the screw.
- Check that there are no foreign objects in the bucket, which may cause damage to the feeding screw.
- Remember correct working methods and avoid getting down unnecessarily from the loader.
- Check regularly to ensure there are no hydraulic leaks. Refer to Section 7 for maintenance instructions.
- Ensure that the loader and the attachment are used in a safe and intended manner. Do not allow children to operate the equipment.



Clean the oil cooler core of the loader regularly; refer to the loader manual for more instructions. Overheating of the oil decreases the power and affects the service life of the hydraulic components of both the attachment and the loader.



Always stop the attachment before leaving the driver's seat. Do not go near the attachment, if another person is operating the controls of the loader. Shut down the loader engine and release the residual pressure before reaching into the bucket or to the discharge opening. The attachment may start unexpectedly when removing a blockage, if the hydraulic system is pressurized.



Note the effects of heavy loads to the handling characteristics of the loaders, especially when working with lighter 300-500 –series loaders. When driving, keep the telescopic boom retracted and keep the load close to the ground. Use additional counterweights as needed. Read the operator's manual of the loader.



Avoid hazard of crushing by ensuring that the parking brake of the loader is engaged when loading the bucket manually. Also ensure that the bucket is lowered firmly on the ground.



The bucket can be filled from a pile like a normal bucket. Do not rotate the feeding screw while loading the bucket. Especially when using a bucket with a long base, the correct working method should be observed for the best performance of the equipment. Filling the large bucket by pushing towards a pile of heavy material and then trying to lift or tilt the bucket is not possible with any loader type. The loading is most effective when only the tip of the bucket is pushed into the pile, and then, by simultaneous driving, lifting and tilting the equipment, material is cut into the bucket. Use high engine revolutions and read the instructions for correct use of the drive pedal from the loader manual.

The dispensing bucket is started by turning the auxiliary hydraulics control lever of the loader to its locked position. Operate the dispensing bucket with moderate engine rpm and with one-pump setting of the auxiliary hydraulics. Depending on the operating conditions, loader model and driving speed the rotating speed of the dispensing bucket can be increased as needed.

7. Inspections, maintenance and servicing

Regular maintenance of the equipment includes keeping the attachment clean and monitoring the condition of the feeding screw. The free end of the screw shaft should be lubricated every 10 hours of use.



Ensure that the bucket is properly supported during all cleaning, inspection and maintenance procedures. Lower the attachment on even ground and keep it attached on the loader. All maintenance procedures must be performed with the loader engine shut down and the hydraulic hoses disconnected.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Leave the service for professional maintenance, if unsure how to perform the repair safely.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system of the attachment or the loader.



7.1. Cleaning and lubrication

Clean the bucket regularly to prevent accumulation of dirt, which is more difficult to remove. You may use a pressure washer and mild detergent. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either.

Do not leave the attachment to a position in which water will stay in the bucket. For storage, place the bucket to the position shown in section **Error! Reference source not found..** Touch up the paint where needed and grease the lubrication points to prevent rust damage. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example.

There is a lubricating point at the bearing of the free end of the dispensing screw shaft, to which a small amount of lubricant is added every 5-10 hours of use.



Lubricant must be added approximately every 5 to 10 operating hours. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, ISO 4254-1

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, ISO 4254-1

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, ISO 4254-1

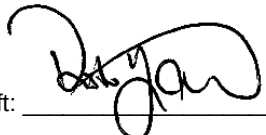
Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:

SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, ISO 4254-1

Kuvaus: Avant-kuormainten kanssa käytettävä hydraulitoiminen jakokauha
 Beskrivning: Hydraulisk utmatareskopa; arbetsredskap för Avant lastare
 Description: Hydraulic dispensing bucket; attachment for Avant loaders
 Beschreibung: Hydraulische Verteil-Schaufel; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant	300/500/600/700 series
Jauhonjakokauha Kraftfoderutmatareskopa Flour/grain dispenser Futtermverteiler	A1194

Allekirjoitus / Namnteckning / Signature / Unterschrift:  10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





AVANT[®]